

Probability Warm-Up: Intersection, Union, and Conditional

Companion handout — complete before the Bayes' Theorem lecture

Name: _____ Date: _____

0. The vocabulary

An **experiment** is anything with an uncertain outcome: rolling a die, testing a patient, drawing a card.

The **sample space** S is the set of all possible outcomes. For one die roll, $S = \{1, 2, 3, 4, 5, 6\}$.

An **event** is a subset of the sample space — something that either happens or doesn't. "The roll is even" is the event $A = \{2, 4, 6\}$.

When every outcome is equally likely:

$$P(A) = \frac{\text{number of outcomes in } A}{\text{number of outcomes in } S}$$

So $P(\text{even}) = 3/6 = 0.5$.

Two facts we will lean on constantly:

- $0 \leq P(A) \leq 1$ always.
- $P(\bar{A}) = 1 - P(A)$, where $\bar{A}$ ("not A ") is the **complement** of A .

Quick check. A single die is rolled. Write out the event $B =$ "the roll is greater than 4" and compute $P(B)$ and $P(\bar{B})$.

$B = \{\text{_____}\}$ $P(B) = \text{_____}$ $P(\bar{B}) = \text{_____}$

1. Intersection — "and"

$A \cap B$ is the event that **both** A and B happen. Read the symbol as "and."

Example. Draw one card from a standard 52-card deck. Let

- $A =$ the card is a heart (13 cards)
- $B =$ the card is a face card — J, Q, K (12 cards)

$A \cap B =$ the card is a *heart face card*. There are exactly 3 of those (J, Q, K), so

$$P(A \cap B) = \frac{3}{52} \approx 0.058$$

Notice that $P(A \cap B)$ is smaller than both $P(A) = 13/52$ and $P(B) = 12/52$. Adding a requirement can only shrink an event — it never grows it. This is worth saying out loud, because it is the sanity check that catches most arithmetic errors later.

If $A \cap B$ is empty — the two events cannot co-occur — we call them **mutually exclusive** and $P(A \cap B) = 0$. “The card is a heart” and “the card is a spade” are mutually exclusive.

Exercise 1. One card is drawn. Let A = “the card is red” and B = “the card is an ace.”

- (a) How many cards are in $A \cap B$? _____
- (b) $P(A \cap B) =$ _____
- (c) Name two events in this deck that are mutually exclusive.

2. Union — “or”

$A \cup B$ is the event that A happens, or B happens, or both. Read the symbol as “or.”

The tempting move is to write $P(A \cup B) = P(A) + P(B)$. That is wrong whenever the events overlap, because everything in the overlap gets counted twice — once in $P(A)$ and again in $P(B)$. The fix is to subtract the double-count:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Example. Same deck, A = heart, B = face card.

$$P(A \cup B) = \frac{13}{52} + \frac{12}{52} - \frac{3}{52} = \frac{22}{52} \approx 0.423$$

Sanity check by counting directly: 13 hearts, plus the 9 face cards that aren’t hearts, is 22 cards.

When the events *are* mutually exclusive, $P(A \cap B) = 0$ and the formula collapses back to plain addition. That’s the special case, not the general rule.

Exercise 2. In a class of 40 students, 22 have written Python before, 15 have taken a statistics course, and 9 have done both.

- (a) How many have done Python *or* statistics (or both)? _____
- (b) $P(\text{Python} \cup \text{Stats}) =$ _____
- (c) How many have done *neither*? _____
- (d) Sketch a Venn diagram and label all four regions with counts.

3. Conditional probability — “given”

$P(A | B)$ is the probability that A happens **given that we already know B happened**. Read the bar as “given.”

Conditioning on B means throwing away every outcome where B didn’t happen. B becomes the new sample space. So we count the outcomes where both happened, and divide by the outcomes where B happened:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

Example. Draw a card. What is $P(\text{face card} | \text{heart})$?

$$P(B | A) = \frac{P(A \cap B)}{P(A)} = \frac{3/52}{13/52} = \frac{3}{13} \approx 0.231$$

Or just reason it out: we know it’s a heart, so there are 13 possibilities, and 3 of them are face cards. Same answer, and the formula is only bookkeeping for that intuition.

Two things to notice

Order matters. $P(A | B)$ and $P(B | A)$ are different questions with different denominators. Above, $P(\text{face} | \text{heart}) = 3/13$, but $P(\text{heart} | \text{face}) = \frac{3/52}{12/52} = 3/12 = 1/4$. Same numerator, different denominator, different answer. **Confusing these two is the single most common mistake in all of probability**, and the entire point of Bayes’ Theorem is to convert one into the other correctly.

Sometimes knowing B tells you nothing. If $P(A | B) = P(A)$, the events are **independent** and the definition rearranges into $P(A \cap B) = P(A)P(B)$. Two coin flips are independent. A card being a heart and being a face card are also independent — check it: $3/13 = P(\text{face} | \text{heart})$ and $12/52 = 3/13 = P(\text{face})$. Equal, so independent. But an infection and a test result had better *not* be independent, or the test is useless.

Exercise 3. Roll two fair dice. Let A = “the sum is 8” and B = “the first die shows 5.”

- (a) $P(A) =$ _____ (there are 36 equally likely outcomes)
- (b) $P(A \cap B) =$ _____
- (c) $P(A | B) =$ _____
- (d) Is $P(A | B)$ bigger or smaller than $P(A)$? What does that tell you?
- (e) Compute $P(B | A)$ and confirm it differs from $P(A | B)$.

4. Putting it together: the two-way table

Most real problems arrive as counts in a table, and every quantity above can be read straight off it.

Out of **1,000 people**, 1% carry an infection. A test is applied to all of them; among the infected it comes back positive 97% of the time, and among the healthy it also comes back positive 3% of the time.

	Test +	Test -	Total
Infected	9.7	0.3	10
Not infected	29.7	960.3	990
Total	39.4	960.6	1,000

(Decimal people are fine — these are expected counts, not a headcount.)

Reading the table:

- **Intersection:** $P(I \cap +) = 9.7/1000 = 0.0097$ — one cell, over the grand total.
- **Marginal:** $P(+) = 39.4/1000 = 0.0394$ — a row or column total, over the grand total.
- **Conditional:** $P(+ | I) = 9.7/10 = 0.97$ — one cell, over its *row* total. Conditioning on I means the “Infected” row is now the whole world.
- **The other conditional:** $P(I | +) = 9.7/39.4 \approx 0.246$ — the same cell, over its *column* total.

Exercise 4. Using the table above:

- $P(\bar{I} \cap +) =$ _____
- $P(- | \bar{I}) =$ _____
- $P(I \cup +) =$ _____ (use the union formula, then verify by counting cells)
- Are I and $+$ independent? Justify with numbers.

Exercise 5 — the punchline. Compare your answers for $P(+ | I) = 0.97$ and $P(I | +) \approx 0.246$.

Both describe “the test and the infection agreeing.” In one sentence, explain to a classmate why they are so far apart. Which of the two would a patient actually want to know?

Answer Key

Quick check. $B = \{5, 6\}$, $P(B) = 2/6 = 1/3$, $P(\bar{B}) = 2/3$.

Exercise 1. (a) 2 (A, A). (b) $2/52 = 1/26 \approx 0.038$. (c) e.g. “heart” and “spade,” or “ace” and “king” — any two that can’t hold at once.

Exercise 2. (a) $22 + 15 - 9 = 28$. (b) $28/40 = 0.70$. (c) $40 - 28 = 12$. (d) Python only 13, both 9, Stats only 6, neither 12 — the four must sum to 40.

Exercise 3. (a) $5/36$ — the sum-8 outcomes are $(2,6), (3,5), (4,4), (5,3), (6,2)$. (b) $1/36$ — only $(5,3)$. (c) $\frac{1/36}{6/36} = 1/6$. (d) $1/6 > 5/36$, so learning the first die is a 5 makes a sum of 8 *more* likely; the events are not independent. (e) $P(B | A) = \frac{1/36}{5/36} = 1/5$, which is not $1/6$.

Exercise 4. (a) $29.7/1000 = 0.0297$. (b) $960.3/990 \approx 0.970$. (c) $P(I) + P(+) - P(I \cap +) = 0.01 + 0.0394 - 0.0097 = 0.0397$; by cells, $(9.7 + 0.3 + 29.7)/1000 = 0.0397$. (d) No. $P(+ | I) = 0.97$ but $P(+) = 0.0394$ — wildly different, so the test result depends heavily on infection status. (Good! An independent test would be worthless.)

Exercise 5. The infection is rare, so the 990 healthy people vastly outnumber the 10 infected ones — and 3% of a large group (29.7 false positives) beats 97% of a tiny group (9.7 true positives). A positive result is therefore most often a false alarm. The patient wants $P(I | +)$, but the test’s advertised “97% accuracy” reports $P(+ | I)$. **Bayes’ Theorem is the tool that converts the number we are given into the number we want.**